

8

Symmetry in crystallography

Mathematics possesses not only cold truth but supreme beauty, a beauty cold and austere, like that of sculpture, sublimely pure, and capable of a stern perfection, such as only the greatest art can show.

Bertrand Russell

In this chapter, we will discuss the concept of symmetry in great detail. We will begin with the description of symmetry operations as coordinate transformations, followed by a discussion of the difference between passive and active operators. Then we introduce rotations, and we determine which rotations are compatible with the 14 Bravais lattices. After a discussion of operators of the first (rotation, translation) and second (mirror, inversion) kinds, we will generate combinations of symmetry operators, which will lead to glide planes and screw axes. Along the way, we will also introduce the time-reversal operator and study how it can be combined with the regular symmetry operators. We conclude the chapter with the definition of point symmetry.

8.1

Symmetry of an arbitrary object

Many objects encountered in nature display some form of symmetry, in many cases only an approximate symmetry; e.g. the human body shows an approximate mirror symmetry between the left and right halves, many flowers have five- or seven-fold rotational symmetry, . . . In the following paragraphs, we will discuss the classical theory of symmetry, which is the theory of symmetry transformations of space into itself.

If an object can be (1) rotated, (2) reflected, or (3) displaced, without changing the distances between its material points and so that it comes into self-coincidence, then that object is symmetric. A transformation of the type (1), (2), or (3), or a combination thereof, that preserve distances and bring the object into coincidence is called a *symmetry operation*. It should be clear that *translations* can only be symmetry operations for infinite objects. The word “symmetric” stems from the Greek word for “commensurate.” Note that the identity operator (i.e., not doing anything) is also considered to be a symmetry property; therefore, each object has at least one symmetry property.

Throughout this chapter and many of the following chapters, we will need to write down shorthand notations for all the symmetry operations that we will introduce. We will

8.1 Symmetry of an arbitrary object

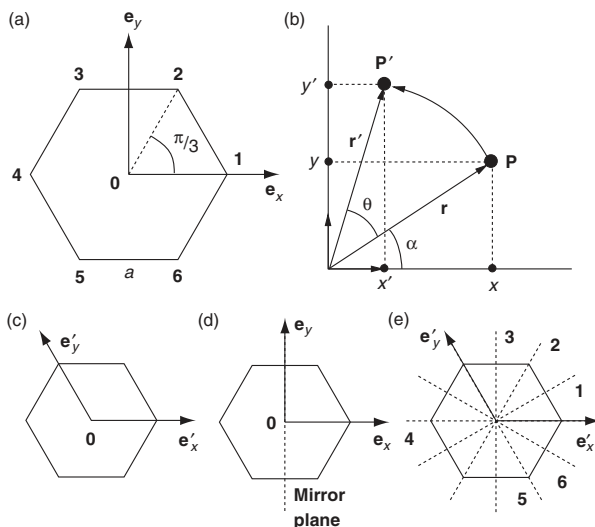


Figure 8.1 Illustration of the six-fold symmetry of the regular hexagon with respect to (a) a Cartesian reference frame and (c) a hexagonal reference frame. In (b), the point $\mathbf{P}$ is rotated by an angle θ to the point $\mathbf{P}'$. (d) and (e) show the presence of mirror planes, again with respect to Cartesian and hexagonal reference frames.

follow the *International Tables for Crystallography, Volume A* (Hahn, 1996) for all notational conventions. In particular, there are two major schools of notation: the *international notation* (also known as the *Hermann–Mauguin notation*) is the standard, which we will follow throughout this text. An alternative notational convention, used primarily by physicists and chemists, is the *Schoenflies notation*. Since this latter notation is widely used in the scientific literature, we will, whenever possible, always give both notations. The Hermann–Mauguin notation will always be used first, followed by the Schoenflies notation in parenthesis; for instance, the identity operator mentioned in the previous paragraph will be denoted by the symbol 1 (E), where 1 is the Hermann–Mauguin symbol and E the Schoenflies symbol. Note that we use a sans-serif font for all operator symbols. While this dual notation is longer than either one used individually, the advantage is that the reader will become familiar with both notational schemes. Most symmetry operations can also be represented by a graphical symbol, and we will introduce those symbols at the appropriate locations throughout the text.

Before we begin a detailed overview of symmetry operations, let us first approach the problem with some simple 2-D considerations. Consider a hexagon with an edge length a , centered in the origin of a Cartesian reference frame, as shown in Fig. 8.1(a). The hexagon has six vertices (corners), labeled 1 through 6. It is obvious that a rotation of 60° around the origin, or any multiple of 60° , will generate a new hexagon that coincides with the first one (i.e., it cannot be distinguished from the first one). According to our

definition of symmetry, this means that the hexagon has rotational symmetry around an axis normal to the hexagon, passing through the origin. If we consider one of the vertices, say point 1, then we can ask the question: what will be the location of this point after we carry out a (counterclockwise) rotation of 60° ?¹ From the drawing, it is clear that point 1 will move to point 2, 2 will move to 3 and so on; the final point 6 will move to 1. In the mathematical literature this can be written as a permutation matrix of the vertices: $(123456) \rightarrow (612345)$. The coordinates of each point can be read from the drawing, but what we really need is a mathematical method to *compute* the coordinates, starting from the initial coordinates.

Consider the 2-D Cartesian reference frame of Fig. 8.1(b). A general point $\mathbf{P}$ has coordinates (x, y) , and is represented by the position vector $\mathbf{r}$. We apply a counterclockwise rotation around an axis perpendicular to the plane of the drawing and going through the origin O . The rotation angle is θ . The point $\mathbf{P}$ is now rotated to the position $\mathbf{P}'$, with position vector $\mathbf{r}'$ and components (x', y') . The angle between the original position vector $\mathbf{r}$ and the $\mathbf{e}_x$ -axis is α . From the drawing, we find that the components of the new position vector are given by:

$$\begin{aligned}x' &= r' \cos(\theta + \alpha), \\y' &= r' \sin(\theta + \alpha),\end{aligned}$$

where $r' = |\mathbf{r}'| = |\mathbf{r}| = r$. Applying the addition theorems for trigonometric functions we find:

$$\begin{aligned}x' &= r(\cos \theta \cos \alpha - \sin \theta \sin \alpha), \\y' &= r(\sin \theta \cos \alpha + \cos \theta \sin \alpha).\end{aligned}$$

From the drawing, we also find that $x = r \cos \alpha$ and $y = r \sin \alpha$, which leads to

$$\begin{aligned}x' &= x \cos \theta - y \sin \theta, \\y' &= x \sin \theta + y \cos \theta,\end{aligned}$$

which can be rewritten in matrix form as:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (8.1)$$

We find that a rotation around an axis can be represented by a *rotation matrix* $D_2(\theta)$. In two dimensions, we have:

$$D_2(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

¹ It is important to note that a counterclockwise rotation is the common and international crystallographic convention for defining a rotation operation. In literature from other fields, the opposite clockwise rotation has been chosen as the convention. In reading the literature it is, therefore, important to identify the convention being used.

8.1 Symmetry of an arbitrary object

In 3-D, the rotation matrix will become a 3×3 matrix. The easiest case is the one that we already worked out, since the third dimension $\mathbf{e}_z$ can be chosen along the rotation axis. The rotation axis itself does not change during the rotation, and we say that the rotation axis is *invariant*. The 3-D rotation matrix is therefore given by

$$D_3(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (8.2)$$

For simplicity, we will, throughout this book, drop the subscript 3 on D_3 , i.e., $D \equiv D_3$. The coordinate transformation in 3-D is explicitly given by:

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad (8.3)$$

or, alternatively:

$$\begin{aligned} x' &= x \cos \theta - y \sin \theta, \\ y' &= x \sin \theta + y \cos \theta, \\ z' &= z. \end{aligned}$$

As expected, we find that the third component z does not change during the rotation.

Now that we have an expression for the coordinate transformation matrix $D_2(\theta)$, we can determine the coordinates of the rotated points in Fig. 8.1(a) for $\theta = \pi/3$ (i.e., 60°). The matrix is given by

$$D_2\left(\frac{\pi}{3}\right) = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix},$$

so that the point 1 with coordinates $(a, 0)$ is rotated into the point with coordinates $a(1/2, \sqrt{3}/2)$, which corresponds to point 2 (Fig. 8.1(a)).

Rotation matrices have a number of special properties:

- (i) *The inverse of a rotation matrix is equal to the transpose of that matrix.* This property makes it extremely easy to compute the inverse of any rotation matrix. Any matrix whose inverse is equal to its transpose is generally known as an *orthonormal* matrix.

As an example, consider the 2×2 rotation matrix $D_2(\pi/4)$:

$$D_2\left(\frac{\pi}{4}\right) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The transpose of this matrix is

$$D_2^T\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Upon multiplication of $D_2(\pi/4)$ with its transpose we find:

$$\frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which proves that $D_2(\pi/4)$ is an orthonormal matrix.

- (ii) *For an orthonormal matrix, the sum of the squares of all elements on any row or column is equal to 1.* This can easily be seen from Eq. 8.1, since $\cos^2 \theta + \sin^2 \theta = 1$.
- (iii) *The determinant of a rotation matrix is always equal to +1.* It is easy to show that $\det(D_2(\theta)) = \det(D(\theta)) = \cos^2 \theta + \sin^2 \theta = 1$.

At this point it is useful to connect what we have just derived with the theory of coordinate transformations presented in the previous chapter (Section 7.4 on page 135). In that section, we defined a transformation matrix, α_{ij} , that transforms the old basis vectors, $\mathbf{a}_j$, into the new ones, $\mathbf{a}'_i$, as follows: $\mathbf{a}'_i = \alpha_{ij}\mathbf{a}_j$. For the components of vectors, we found, in Table 7.1, that the transformation relation is given by:

$$p'_i = p_j \alpha_{ji}^{-1};$$

in other words, to transform the coordinates, p_j , of a point, we must post-multiply by the transpose of the inverse of α_{ij} . In the present section, Eq. (8.3) can be rewritten as: $p'_i = D_{ij}p_j$. These two relations describe the same coordinate transformation, so that we must have:

$$D = \alpha^{-1}. \quad (8.4)$$

This means that we have two equivalent descriptions for the coordinate transformation (in this case a counterclockwise rotation by $\pi/3$): the description with the matrix $D(\theta)$ operates directly on the coordinates p_i , whereas the description with the matrix α operates on the basis vectors $\mathbf{a}_j$. Both descriptions describe the same transformation, and we will return to the distinction between them in Section 8.2 on page 158.

In all examples above, we have worked with the Cartesian reference frame ($\mathbf{e}_x, \mathbf{e}_y$) defined in Fig. 8.1(b). As a consequence of this choice, the entries of the rotation matrices are typically non-integer numbers, such as $1/2$ or $\sqrt{3}/2$. If we select a different reference frame, for instance ($\mathbf{e}'_x, \mathbf{e}'_y$) as shown in Fig. 8.1(c), then the rotation matrix changes to a much simpler form, which we will now derive for the 3-D case.

To determine the entries of the rotation matrix $D(\pi/3)$ with respect to the primed basis vectors, all we need to do is figure out what happens to the basis vectors themselves. For instance, the vector $\mathbf{e}'_x$ is rotated to $\mathbf{e}'_x + \mathbf{e}'_y$, and $\mathbf{e}'_y$ is rotated to $-\mathbf{e}'_x$. The vector $\mathbf{e}'_z$ remains

8.1 Symmetry of an arbitrary object

unchanged. These three relations are sufficient to determine the entire matrix, since we can write:

$$(\mathbf{e}'_x + \mathbf{e}'_y \quad -\mathbf{e}'_x \quad \mathbf{e}'_z) = (\mathbf{e}'_x \quad \mathbf{e}'_y \quad \mathbf{e}'_z) \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The matrix on the right-hand side is the rotation matrix, D , with respect to the primed reference frame. We see that it contains only the integers 0, -1 , and $+1$. It is straightforward to show that the points 2 through 6 of Fig. 8.1(a) can be obtained by repeated application of the matrix D on the initial point 1. The standard procedure for the determination of a transformation matrix then consists of two steps:

- (i) Determine from a drawing how each of the basis vectors transforms under the symmetry operation, and write the transformed basis vector as a linear combination of the untransformed basis vectors;
- (ii) Take the coefficients of this linear combination and write them as the columns of the matrix D . The coefficients of the first transformed basis vector go in the first column, and so on.

Let us consider another example: a mirror plane containing the Cartesian $\mathbf{e}_y$ and $\mathbf{e}_z$ basis vectors (dashed line in Fig. 8.1(d)). This mirror plane leaves the vectors in the plane unchanged, and replaces $\mathbf{e}_x$ by $-\mathbf{e}_x$. Therefore, the transformation matrix is given by:

$$D(m) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where the argument m of the matrix represents the mirror plane. We will introduce a more complete notation in a later section. Note that this matrix has a negative determinant, equal to -1 . This indicates that this symmetry operation results in a reversal of handedness, which we will discuss in detail in Section 8.2.5. The entries in this matrix are simple integers, because the mirror plane contains two of the basis directions. It is left as an exercise for the reader to determine what the matrix would look like if the mirror plane were rotated clockwise by 30° .²

If we describe the mirror planes of the hexagon (all six of them are shown in Fig. 8.1(e)) with respect to the primed basis vectors, then once again all mirror planes can be represented by matrices that contain only the entries 0, -1 , and $+1$. The matrices for the mirrors 1 and 5 are given by:

$$D(m_1) = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D(m_5) = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

² The entries in this new matrix will no longer be simple integers.

We leave it to the reader to determine the remaining four matrices $D(m_{2,3,4,6})$. In the remainder of this book, we will nearly always use symmetry transformation matrices that are defined with respect to the basis vectors of the Bravais unit cell, so that the entries of those matrices will be the integers 0, -1 , and $+1$. This will allow us to simplify the description of the symmetry of all crystals. In the following section, we will discuss in more detail the nature of symmetry operations, in particular those that are relevant to crystallography.

8.2

Symmetry operations

There are two ways to view the action of a symmetry operator on an object: one can either interpret the symmetry operator as *acting on the object*, leaving the reference frame unchanged in space or, equivalently, one can let the operator *act on the reference frame*, leaving the object unchanged in space. If we represent the object by the symbol $\mathcal{F}$, and the vector $\mathbf{r}$ represents a point of the object, then this relation can be expressed mathematically as follows:

$$\mathcal{F}(\mathcal{O}[\mathbf{r}]) = \mathcal{O}^{-1}[\mathcal{F}(\mathbf{r})], \quad (8.5)$$

where $\mathcal{O}$ represents the symmetry operator and $\mathcal{O}^{-1}$ its inverse. This can be understood easily when the operator is a simple rotation: rotation of an object through $+60^\circ$ is the same as rotating the reference frame through an angle of -60° (see Fig. 8.2). If the object is moved in space, leaving the reference frame constant, then the operator is said to be an *active operator*. If the reference frame is moved in space, leaving the object unchanged, then the operator is said to be a *passive operator*. In this text, we will always consider a symmetry operator as acting on the coordinates of the material points, rather than on the reference frame. In other words, we will always take the *active* interpretation of a symmetry operation.

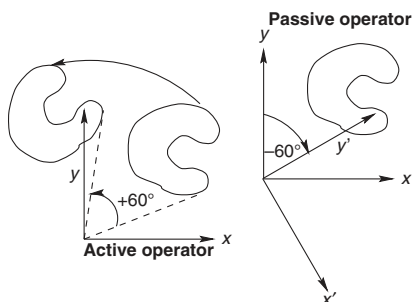


Figure 8.2 Illustration of an *active* and a *passive* operator.

8.2 Symmetry operations

We saw in the previous section, that there are two equivalent matrix descriptions for the coordinate transformation corresponding to the symmetry operation: the matrix $D(\theta)$ transforms the coordinates, p_i , and is, hence, a representation of the *active* interpretation. The other matrix, α , transforms the basis vectors, and is, hence, a representation of the *passive* interpretation.

Any physical property of the object or crystal will be invariant under the symmetry operator:

$$\mathcal{F}(\mathbf{r}) = \mathcal{F}(\mathcal{O}[\mathbf{r}]) = \mathcal{F}(\mathbf{r}'). \quad (8.6)$$

This implies that the inverse transformation, from $\mathbf{r}'$ to $\mathbf{r}$ exists and is also a symmetry operation. If several symmetry properties exist, then it can be shown that any combination of them will also be a symmetry property. We will study such combinations of symmetry operators in the next chapter, when we introduce the concept of *point group*.

Symmetry is an inherent attribute of an object; without a certain external measure, symmetric orientations of an object are indistinguishable. Note that such an external measure would destroy the symmetry because it would require the attachment of a reference point to the object. Hence, symmetry can be a very subtle aspect of an object; the more intricate symmetry properties, such as time-reversal symmetry in magnetic materials, can be difficult to detect and understand. For now, we will restrict ourselves to study basic symmetry operations, namely the isometric transformations of an object.

8.2.1

Basic isometric transformations

An isometric transformation is a transformation that leaves the metric properties of space unaltered, i.e., no stretching, twisting, or bending is involved. Obviously, empty space is invariant under any kind of symmetry operator, even the non-isometric ones. To distinguish between isometric and non-isometric operators, we have to provide a gage by which distances and coordinates can be measured. In 3-D space, this can be done by providing three basis vectors or four non-coplanar points, i.e., an *asymmetric tetrahedron*. The four points are the origin and the three endpoints of the basis vectors. Any fifth point can then be located with respect to this tetrahedron and, hence, any object can be fully and unambiguously defined, except for its *symmetric variants*.

It can be shown that *any isometric transformation can be reduced to either a translation, a rotation, or a reflection, or a combination of these*. The mathematical proof uses the concept of the *Chasles-center* but this is beyond the scope of this text. The proof involves consideration of two identical asymmetric tetrahedra in different orientations on different locations and determination of the ways to bring them into self-coincidence.

There are two different correspondences between an object and its symmetric variant(s):

- *Coincidental or congruent equality* = a transformation of the first kind;
- *Mirror equality* = a transformation of the second kind.

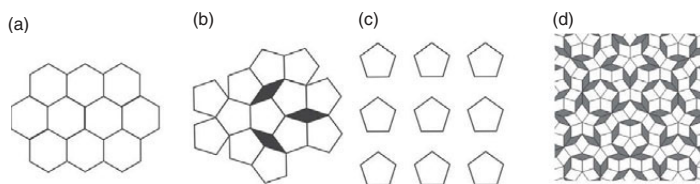


Figure 8.3 (a) Tiling of the 2-D plane with hexagons illustrating the compatibility of a six-fold symmetry axis and crystalline periodicity; (b) illustration of the fact that the 2-D plane cannot be tiled with pentagons; (c) square lattice decorated with pentagons; and (d) 2-D Penrose tiling, which preserves five-fold rotational symmetry.

Transformations of the first kind are also known as *proper motions*, since they can be physically realized.³ The mirror operation is not a proper motion, because it is physically impossible to change the handedness of an object without actually deforming it. Therefore, this type of operation is distinguished from the proper motions. Any transformation of the first kind is either a *translation* or a *rotation* or a combination of both. A transformation of the second kind is either a *reflection* or an *inversion*. Before discussing the various fundamental symmetry operations in more detail, we will first derive some important constraints on the crystallographically permitted rotational symmetries.

8.2.2

Compatibility of rotational symmetries with crystalline translational periodicity

In this section, we address the question of compatibility (or incompatibility) of rotational symmetry axes with the translational symmetry of crystalline solids. As an example, consider the hexagonal tiling of the 2-D plane in Fig. 8.3(a). There are no gaps in between the tiles, so that no area is left uncovered. If we attempt to repeat this with a five-fold tile, then we quickly find (Fig. 8.3(b)) that we cannot tile the 2-D plane with tiles of five-fold symmetry. The same is true of higher-order polygons, such as the heptagon, the octagon, and so on. Figure 8.3(c) shows an example of what may be a *molecular solid* in which pentagonal molecules decorate the sites of a square lattice. Although the local units are pentagons, the five-fold symmetry axis is lost since it is inconsistent with the square lattice. Furthermore, the four-fold symmetry, which would exist when spherical atoms decorate the square lattice, is lost by virtue of the lack of four-fold symmetry in the pentagonal basis. Figure 8.3(d) shows an example of what may be called a *2-D quasicrystal*, in which two different rhombuses are used to fill the plane, in a such a way that five-fold rotational symmetry is preserved. In Chapter 16, we will discuss in more detail quasi-periodic motifs which tile the plane and preserve five-fold rotational symmetries but do not involve tilings with pentagons. This notion can be generalized to

³ We all know from experience that we can translate and rotate arbitrary objects, but we cannot change their handedness without temporarily changing their shape. For instance, to turn a left-handed glove into a right-handed glove, we must turn it inside-out, which is not an isometric operation.

8.2 Symmetry operations

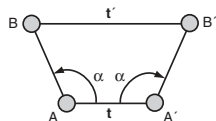


Figure 8.4 Lattice translation vector, $\mathbf{t}$, connecting points A and A'. The lattice points B and B' are generated by a rotational symmetry axis (rotation by α) normal to the plane and passing through sites A or A', respectively.

include quasiperiodic tilings preserving other symmetries, either non-crystallographic or crystallographic.

The fact that only one-, two-, three-, four-, and six-fold axes are compatible with crystalline axes was elegantly demonstrated using the *rule of rational indices* (Haüy, 1784, 1801, 1822). The rule of rational indices is illustrated geometrically in Fig. 8.4. Consider the two lattice points A and A', which are separated by a unit translation vector, $\mathbf{t}$. Consider also a rotation operator, D , with an axis normal to the plane containing the vector AA' and passing through the point A. If D represents a counterclockwise rotation by the angle α , it takes lattice point A' into a symmetrically equivalent lattice point B. Since every rotational symmetry operation has an inverse operation, we may also consider a rotation by $-\alpha$ about the point A', which yields the lattice point B' starting from A. Finally, if the rotations are consistent with the translational symmetry of the lattice, then the distance t' (the length of $\mathbf{t}'$) between B and B' must be an integral number of unit translations t (the length of $\mathbf{t}$), i.e.:

$$t' = mt, \quad (8.7)$$

where m is an integer. From examination of the geometry shown in Fig. 8.4, it is also clear that

$$t' = t + 2t \cos \alpha. \quad (8.8)$$

Combining the previous equations, we conclude that

$$\cos \alpha = \frac{1 - m}{2}. \quad (8.9)$$

If m is an integer, then so is $(1 - m)$. Furthermore, we know that $|\cos \alpha| \leq 1$, and, therefore, $|1 - m| \leq 2$. This, in turn, means that $1 - m = -2, -1, 0, 1, 2$ and, hence, $\alpha = \pi, 2\pi/3, \pi/2, \pi/3$, or 0, respectively. Thus, the only rotations that are consistent with the translational symmetry found in periodic crystals are the one-fold ($\alpha = 2\pi$), two-fold ($\alpha = \pi$), three-fold ($\alpha = 2\pi/3$), four-fold ($\alpha = \pi/2$), and six-fold ($\alpha = \pi/3$) rotations. Note in particular the absence of five-fold rotations; in Chapter 16, we will describe how crystals with five-fold rotational symmetry *but no translational symmetry* can be constructed.

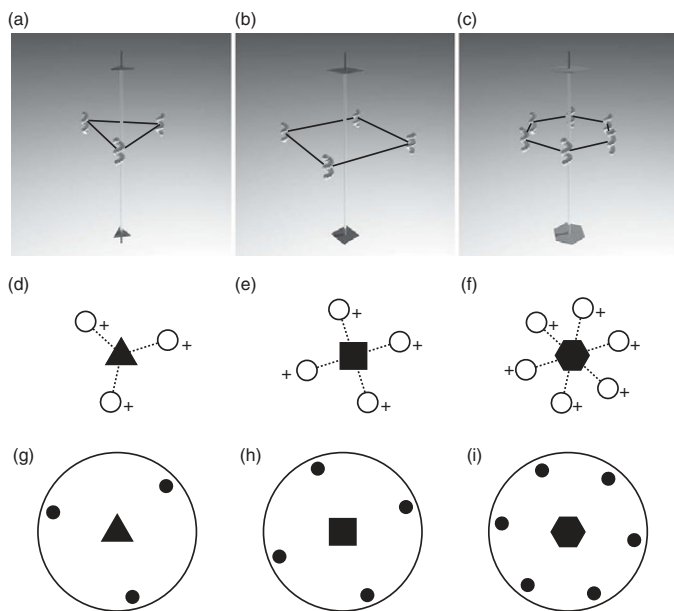


Figure 8.5 Rendered 3-D drawings of (a) three-fold, (b) four-fold, and (c) six-fold rotational symmetry, along with 2-D drawings of the symmetry operator (second row), and the corresponding stereographic projections.

8.2.3

Operations of the first kind: pure rotations

A pure rotation is characterized by a *rotation axis* and a *rotation angle* that is chosen to be positive for a counterclockwise rotation. The rotation axis can be described by a direction vector $[uvw]$, or by the equation of the line that represents the rotation axis. It is customary to write the rotation angle as a fraction of 2π , i.e., $\alpha = 2\pi/n$. The number n is the *order* of the rotation and we say that a rotation is n -fold if its angle is given by $2\pi/n$. The order n can take any integer value from $n = 1$ (the identity operator, 1 (E)), to $n = \infty$ for a circle. Examples of rotation axes of orders 3, 4, and 6 are shown in Fig. 8.5. Note that the object that is used to illustrate the rotational symmetry, in this case a distorted helical string of spheres, should always be taken to be an object that itself does not have any symmetry other than the identity operator. If we had used a sphere as object, we would not have discovered that the orientation of the object changes while it is rotated.

A pure rotation is denoted by the symbol $n (C_n)$, with n the order of the rotation.⁴ In many cases, the rotation axis will be parallel to the third basis direction, i.e., parallel to the

⁴ Recall that we state the Hermann–Mauguin symbol first, followed by the Schoenflies symbol between parentheses.

8.2 Symmetry operations

crystallographic **c**-axis. In those cases, we do not need to state explicitly the orientation of the rotation axis. For all other cases, however, the rotation symbol must be augmented by information about the orientation of the rotation axis. This can be done in two ways: we can use the direction indices $[uvw]$ of the rotation axis, or we can use the equation of the line coinciding with the axis. Hence, a three-fold rotation axis aligned along the $[111]$ direction of a cubic reference frame can be represented as $3(C_3)[111]$. The equation of the line along this direction is given by $x = y = z$, so that an alternative notation would be $3(C_3)x, x, x$. This latter notation is the one used in the *International Tables for Crystallography* (Hahn, 1996); the former notation is used in many textbooks, e.g., Burns and Glazer (1990).

As another example, consider a two-fold rotation axis, oriented so that it is normal to the **c**-axis of a cubic reference frame, and bisects the angle between the **a** and **b** directions. What are the two full symbols for this rotation operator? The two-fold rotation axis is represented by the symbol $2(C_2)$, but since it is not oriented along the **c**-axis, we must add information about the orientation. It is easy to verify that the axis lies along the $[110]$ direction, so that the symbol is written as $2(C_2)[110]$. The equation for the line along this direction is given by $x = y$ and $z = 0$, so that the international symbol is given by $2(C_2)x, x, 0$.

In drawings, we use a standard symbol to indicate an n -fold rotation axis. The symbol is a filled regular polygon with n sides. A six-fold rotation axis perpendicular to the drawing plane is then indicated by the $\bullet$ symbol, a three-fold axis by $\blacktriangle$, and a two-fold axis by $\bullet$. The use of these symbols is illustrated in Fig. 8.5(d) through (i); the center row shows the graphical symbol surrounded by open circles with a plus sign next to them. The rotation operator takes any one of these points and rotates it onto the next one (in a counterclockwise sense). During the rotation, the elevation of the point does not change, so that if one point is at some positive distance above the plane of the drawing, indicated by the $+$ sign, then all of them are. This is consistent with the fact that rotations occur in a plane normal to the rotation axis. The bottom row of Fig. 8.5 shows a commonly used representation of symmetry operations by means of stereographic projections. The rotation axis is aligned along the north–south center line, which is normal to the plane of the drawing, and a point in the northern hemisphere, represented by a filled circle, is rotated to its equivalent positions, all of them in the northern hemisphere. In the remainder of this book, we will make frequent use of both graphical representation schemes.

In Section 8.1, we have seen that rotations can be represented by orthogonal matrices $D(\theta)$, and we considered examples of rotations around the z -axis of the reference frame. Additional examples are shown in Fig. 8.6. Figure 8.6(a) shows a four-fold rotation around

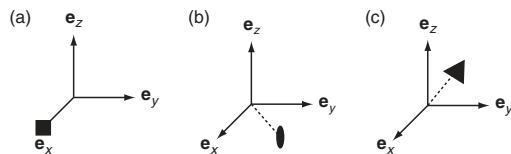


Figure 8.6 (a) Four-fold, (b) two-fold, and (c) three-fold rotations used as examples in the text.

the x -axis of a cubic reference frame. The $\mathbf{e}_x$ basis vector is obviously invariant, whereas $\mathbf{e}_y$ is rotated onto $\mathbf{e}_z$, and $\mathbf{e}_z$ is rotated to $-\mathbf{e}_y$. We find for the transformation matrix:

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (8.10)$$

For a two-fold rotation $2 (C_2) x, x, 0$ (or $2 (C_2) [110]$), shown in Fig. 8.6(b), we find

$$D = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (8.11)$$

and for a $3 (C_3) x, x, x$ rotation ($3 (C_3) [111]$, Fig. 8.6(c)) we have

$$D = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (8.12)$$

8.2.4

Operations of the first kind: pure translations

A pure translation is characterized by a translation vector $\mathbf{t}$. We have already discussed translations in Chapter 4. In crystals, a translation vector is limited to the lattice vectors and, if present, the lattice centering vectors. We will see later that certain fractions of lattice translation vectors may also be allowed. In drawings, translation vectors are indicated by arrowed lines, i.e., $\longrightarrow$. The international symbol for a translation is $t(u, v, w)$, as in $t(1/2, 0, 1/2)$; alternatively, one could also use $t[uvw]$, since a translation implies a direction, which is described by a direction symbol. We remind the reader that a translation is only a true symmetry operation in an infinite solid.

In the preceding sections, we have seen that rotations can be represented mathematically by matrices. In the remainder of this section, we will explore the possibility of representing translations by means of matrices. This may seem a bit odd, since a translation is usually represented as a vector addition, e.g., translation by a vector $\mathbf{t}$ is described by the equation:

$$\mathbf{r}' = \mathbf{r} + \mathbf{t}. \quad (8.13)$$

It is clear from this relation that we cannot simply multiply $\mathbf{r}$ by a 3×3 matrix to get the same result. It would be useful to have a matrix representation for a translation, since then we could represent *all* symmetry operations by matrices. It turns out that we can do this by working with 4-D vectors instead of the regular 3-D vectors. This sounds complicated but it is really very simple: to each triplet of vector components (x_1, x_2, x_3) we simply add the number 1 as fourth component, i.e., $(x_1, x_2, x_3, 1)$. These types of coordinates are called

8.2 Symmetry operations

normal coordinates or homogeneous coordinates.⁵ The fourth component *must always be equal to 1*, and is denoted by x_4 .

Consider the component notation for Eq. 8.13:

$$x'_i = x_i + u_i,$$

where the components of $\mathbf{t}$ are denoted by u_i . This relation is valid when the index i takes on the values 1, 2, and 3. If we assume that it is also valid when the index, i , goes from 1 to 4, then we can write out the following explicit relations (recall that $x_4 = u_4 = 1$):

$$\begin{aligned} x'_1 &= 1 \times x_1 + 0 \times x_2 + 0 \times x_3 + u_1 \times x_4, \\ x'_2 &= 0 \times x_1 + 1 \times x_2 + 0 \times x_3 + u_2 \times x_4, \\ x'_3 &= 0 \times x_1 + 0 \times x_2 + 1 \times x_3 + u_3 \times x_4, \\ x'_4 &= 0 \times x_1 + 0 \times x_2 + 0 \times x_3 + u_4 \times x_4. \end{aligned}$$

The last equation is trivial, since it simply states that $1 = 1$. We can rewrite this set of equations in matrix form, as follows:

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & u_1 \\ 0 & 1 & 0 & u_2 \\ 0 & 0 & 1 & u_3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{pmatrix}. \quad (8.14)$$

It is clear that this equation is completely equivalent to $\mathbf{r}' = \mathbf{r} + \mathbf{t}$, because when we write it out in components, we recover the correct relation. By using 4-D vectors instead of 3-D, we have been able to include the translation components u_i as part of a 4×4 matrix. We don't have to pay any attention to the last equation of this set of four, since it is a trivial one ($1 = 1$).⁶

We will introduce the symbol $\mathscr{W}$ for this 4×4 matrix. In other words

$$\tilde{x}'_i = \mathscr{W}_{ij} \tilde{x}_j,$$

⁵ Four-dimensional coordinates are often used in the world of computer graphics (e.g., [Salmon and Slater \(Chapter 13, 1987\)](#)) and make it relatively easy to carry out a large number of different coordinate transformations, such as rotations and translations (which we discuss in more detail in the main text), but also scaling, orthogonal, and perspective projections from 3-D to 2-D. In particular, viewing transformations, which convert 3-D objects into 2-D representations on a computer screen, are conveniently expressed in normal coordinates. A general homogeneous coordinate corresponding to a Cartesian point (x, y, z) is described by (wx, wy, wz, w) , with $w \neq 0$. Equivalently, the 4-D coordinate (x, y, z, w) corresponds to the point with Cartesian coordinates $(x/w, y/w, z/w)$. In all cases of importance for crystallography, we may take $w = 1$, so that conversion between conventional and homogeneous coordinates is simply a matter of dropping (or adding) the fourth component $x_4 = 1$.

⁶ It is good practice always to carry out all computations including this fourth row; at the end of a computation this row should always consist of three zeros and a one. When this is not the case, then there must have been a computational error somewhere!

where the tilde indicates that normal coordinates are being used. The indices i and j both take on values from 1 to 4. Note that $\mathcal{W}$ consists of three parts: a 3×3 matrix in the upper left corner, a 3×1 column in the top right-most column, and a 1×4 row at the bottom:

$$\mathcal{W} = \left(\begin{array}{ccc|c} 1 & 0 & 0 & u_1 \\ 0 & 1 & 0 & u_2 \\ 0 & 0 & 1 & u_3 \\ \hline 0 & 0 & 0 & 1 \end{array} \right).$$

Since the bottom row does not contain any information, we can define a new notation, which is equivalent to the matrix $\mathcal{W}$. This notation is known as the *Seitz symbol*, written as $(D|\mathbf{t})$. The symbol D is the 3×3 sub-matrix of $\mathcal{W}$, and $\mathbf{t}$ represents the translation. For a translation, D is the identity matrix, which we can represent by the symbol E . Therefore, the Seitz symbol for a translation is given by $(E|\mathbf{t})$. It is always straightforward to construct the matrix $\mathcal{W}$ when the Seitz symbol is known.

We can also use this notation for rotations. In that case, there is no translation, so that $\mathbf{t} = \mathbf{0}$, and the Seitz symbol for a pure rotation becomes $(D|\mathbf{0})$. The $\mathcal{W}$ matrix for a rotation is then given by:

$$\mathcal{W} = \left(\begin{array}{ccc|c} D_{11} & D_{12} & D_{13} & 0 \\ D_{21} & D_{22} & D_{23} & 0 \\ D_{31} & D_{32} & D_{33} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right). \quad (8.15)$$

The Seitz symbol for the identity operator is simply $(E|\mathbf{0})$.

8.2.5

Operations of the second kind: pure reflections

In Fig. 8.7 the two different types of operation of the second kind are depicted: a pure reflection, and an inversion. A pure *reflection*, also known as a *mirror*, is characterized by a plane, indicated in a drawing by a thick solid line ——— if the plane is at an angle with respect to the plane of the drawing. If the mirror plane lies in the plane of the drawing, then the symbol is $\perp$. Examples are shown in Fig. 8.7(c) and (d). The open circle with a plus sign is copied into the second circle, which now has opposite handedness. This is indicated by means of a comma inside the circle. When the two circles coincide in the drawing, as in (d), then a vertical line divides the circle into two halves, each with its own + or – sign, and the appropriate comma. The corresponding stereographic projections are shown in Fig. 8.7(f) and (g). Note that in stereographic projections, one does not usually distinguish between points of opposite handedness.

The Hermann–Mauguin symbol for a mirror plane is m , while the Schoenflies symbol is the Greek letter σ , resulting in the notation $m(\sigma)$. Sometimes, the Schoenflies symbol has

8.2 Symmetry operations

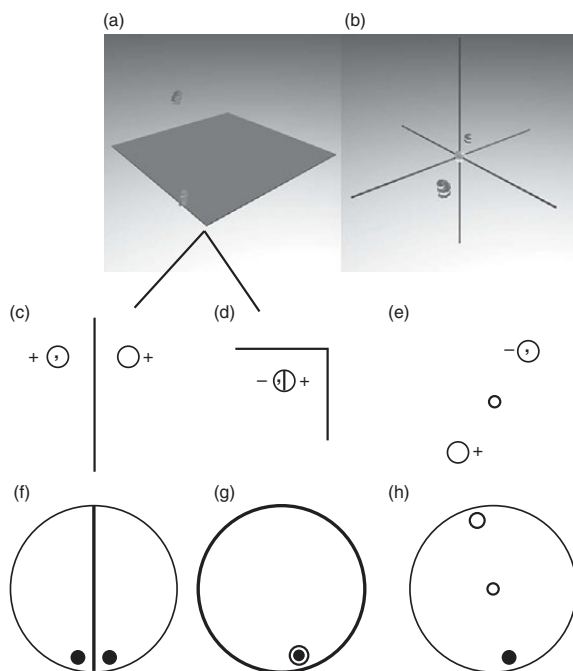


Figure 8.7 The two different types of second kind operation: pure reflection, and inversion center.

a subscript which indicates the orientation of the mirror plane. The symbol $m(\sigma_h)$ refers to a mirror plane that is normal to a given direction (typically a rotation axis), whereas $m(\sigma_v)$ represents a mirror plane which contains that direction. An additional symbol, $m(\sigma_d)$, is also used as a variant of $m(\sigma_v)$.

The international notation for a mirror plane is sometimes extended by providing information on the orientation of the plane. This can again be done in two possible ways, either by specifying the direction of the normal to the mirror plane, or by writing down the equation of the plane. A mirror plane parallel to the (110) plane in a cubic system would be written as either $m(\sigma)[110]$ or $m(\sigma)x, -x, 0$.

We have already seen how we can represent a mirror operation by a matrix of the type $D(m)$ (see page 157). It is then straightforward to write down the Seitz symbol for a mirror operation: $(D(m)|\theta)$.

8.2.6

Operations of the second kind: inversions

An *inversion* is a point symmetry operation that takes all the points $\mathbf{r}$ of an object and projects them into the new points given by $-\mathbf{r}$. The operation is denoted by the symbols $\bar{1}$

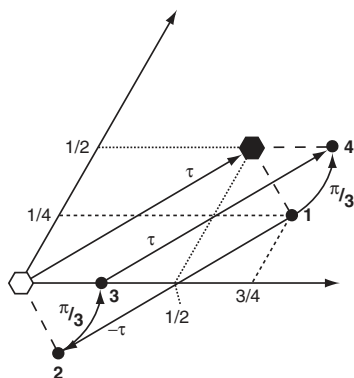


Figure 8.8 Decomposition of a six-fold rotation at location $(1/2, 1/2, 0)$ into two translations and a rotation about a six-fold axis through the origin.

(i). The inversion operation is illustrated in Fig. 8.7(b); the helix in the front of the image is inverted to the back. Note that this changes the handedness of the helix. In drawings, such as Fig. 8.7(e), the inversion center is denoted by the symbol $\odot$ (i.e., a small open circle). It is impossible to bring an inverted object into coincidence with the original object by rotations and translations alone. Inversion symmetry only exists in three dimensions, and is equivalent to a 180° rotation in two dimensions (proof left for the reader). In stereographic projections, the inversion is represented as shown in Fig. 8.7(h).

It is easy to see that the matrix representing the inversion operator is simply $-\mathbf{E}$, i.e., a matrix with -1 along the diagonal and zero elsewhere. The Seitz symbol for the inversion operation is then $(-\mathbf{E}|\mathbf{0})$.

8.2.7

Symmetry operations that do not pass through the origin

All of the operators introduced so far always passed through the origin of the reference frame. Obviously, it is possible for a symmetry operator to be located at some other point (this will become important when we talk about space groups in Chapter 10). We must then ask: how do we compute the matrix $\mathcal{W}$ for a symmetry element that is not located at the origin? As an example, consider the six-fold axis $\bullet$ located at the point $(1/2, 1/2, 0)$ and parallel to the $\mathbf{c}$ direction of the reference frame shown in Fig. 8.8. Rotation of point 1 over an angle of 60° can be decomposed into three elementary steps (elementary in the sense that the decomposition consists only of pure operations, as introduced in the preceding sections). If we call the position vector of the rotation axis $\boldsymbol{\tau}$, then the following three steps are equivalent to the original rotation:

- (i) Translate the point 1 by a vector $-\boldsymbol{\tau}$ to the point 2;
- (ii) Rotate counterclockwise by an angle of 60° around an axis through the origin to the point 3;
- (iii) Translate by a vector $\boldsymbol{\tau}$ to the point 4.

8.3 Magnetic symmetry operations

This is the same as going directly from 1 to 4 by application of the rotation axis at τ . We already know that a translation is described by a matrix $\mathcal{W}_\tau$ of the form:

$$\mathcal{W}_\tau = \begin{pmatrix} 1 & 0 & 0 & \tau_1 \\ 0 & 1 & 0 & \tau_2 \\ 0 & 0 & 1 & \tau_3 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

so that the complete transformation is given by the ordered matrix product (i.e., the right-most operation is carried out first):

$$\begin{aligned} \mathcal{W} &= \mathcal{W}_\tau \mathcal{W}_{\bullet} \mathcal{W}_{-\tau} \\ &= \begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}; \\ &= \begin{pmatrix} 0 & -1 & 0 & 1 \\ 1 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

If we multiply this matrix by a general position vector $(x, y, z, 1)$, then the result is $(1 - y, x + y - 1/2, z, 1)$. For the point 1 in Fig. 8.8, with normal coordinates $(3/4, 1/4, 0, 1)$, we find the new coordinates $(3/4, 1/2, 0, 1)$, which is in agreement with the graphical result. This decomposition technique is generally valid for any symmetry operation that does not pass through the origin.

8.3

*Magnetic symmetry operations

8.3.1

Time-reversal symmetry and axial vectors

In Sections 1.5.4 and 3.7 we have introduced the concept of magnetic symmetry by means of an example, and by showing how the 22 magnetic Bravais lattices can be derived. To do so, we had to introduce symmetry operations that reverse time, in addition to performing their regular function. For instance, we worked with the time-reversed basis vectors $\mathbf{a}'$, $\mathbf{b}'$, and $\mathbf{c}'$, which stand for the normal translation vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, followed by the time-reversal operator. The prime following the symmetry operator symbol indicates that time reversal is implied. In the next section, we will consider the more general case of

combinations of two arbitrary symmetry operators; for now, we limit ourselves to the combination of pure operators with the time-reversal operator, which we will represent by the symbol $1' (E')$.

The concept of time reversal may appear a bit odd at first. After all, in our daily lives we experience a distinct asymmetry between the past and the future. However, a closer look at the basic equations of motion (Newton's laws) reveals that there is, in general, no preference for one direction of time or the other. In other words, from a theoretical point of view, the "arrow of time" can point from past to future, or from future to past, without changing the laws of physics. The act of replacing the time t by $-t$ in an equation is known as *time reversal*. If the equation does not change when this replacement is made, then we say that the equation has time-reversal symmetry. As a simple example, consider Newton's second law of motion:

$$\frac{d\mathbf{p}}{dt} = m \frac{d^2\mathbf{r}}{dt^2} = \mathbf{F}, \quad (8.16)$$

where $\mathbf{p}$ is the momentum, m the mass, and $\mathbf{F}$ the applied force. If we replace t by $-t$, the equation remains unchanged, since time enters as a second-order derivative, which has a quadratic dependence on time. Therefore, Newton's second law of motion has time-reversal symmetry.

We know from basic physics that moving electrical charges generate a magnetic field. In particular, electrons moving around the nucleus of an atom generate tiny electrical currents, which are described by a vector quantity known as the *magnetic moment*.⁷ The magnetic moment is represented by a vector $\boldsymbol{\mu}$; the magnitude of this vector is equal to the current multiplied by the area of the circular orbit of radius r , i.e.,

$$|\boldsymbol{\mu}| = I\pi r^2. \quad (8.17)$$

The units of the magnetic moment vector are Am^2 . The direction of the moment vector is determined by the direction of the current; for a counter-clockwise current, the moment vector points up, whereas for a clockwise current the moment points down.⁸ Since an electrical current represents the number of electrical charges Q passing by a particular point per unit of time,

$$I = \frac{dQ}{dt}, \quad (8.18)$$

it is clear that a reversal of time changes the sign of the current and, hence, reverses the direction of the magnetic moment vector $\boldsymbol{\mu}$. From a symmetry point of view, this means

⁷ In classical physics, the orbit of an electron around the nucleus is assumed to be circular. While orbiting electrons should be described by quantum mechanical equations rather than classical equations, the interpretation of a magnetic moment vector as being determined by an electrical current and an area remains valid in quantum mechanics.

⁸ It is customary to use the *right-hand rule* to determine the orientation of the moment vector. If four fingers of the right hand, excluding the thumb, are curled so that the current flows from the knuckles to the finger tips, then the moment points along the direction of the thumb, normal to the plane of the current.

8.3 Magnetic symmetry operations

that we must distinguish between two classes of symmetry operators, namely those that do not change the direction of the current, and those that do.

Before we do so, it is important that we introduce the concepts of polar and axial vectors. We begin by considering the difference between *scalars* and *pseudo-scalars*. In the previous section we have made a distinction between operators of the first and second kind based on how they act on objects that have handedness. If an object, for instance a sphere or a cube, does not have handedness, then we refer to this object as a scalar object. If we operate on such an object with operators of the second kind (mirror, inversion), then we can replace those operations by combinations of operators of the first kind. For instance, when we take a cube with its center at $\mathbf{r}$, and apply the inversion operator, then we obtain a cube with center at $-\mathbf{r}$. It is not too difficult to see that we can obtain the same result by applying a single translation $-2\mathbf{r}$; we can do this, because the cube does not have handedness. However, if the object does have handedness, then the operators of the second kind cannot be represented by combinations of operators of the first kind.

If we represent the handedness of an object by ± 1 , with $+1$ being a right-handed object, and -1 a left-handed object, then the property of handedness is a *pseudo-scalar*. Operators of the second kind will change the handedness from left- to right-handed or vice versa, whereas operators of the first kind do not affect the handedness. Therefore, it is not possible to transform two objects of opposite handedness into each other, using only operators of the first kind.

Next, we ask the question: what happens when the object that we consider cannot be represented by a scalar quantity, but, instead, requires a vector quantity? An obvious example is the magnetic moment vector, $\boldsymbol{\mu}$. Each vector has two characteristics: a length and a direction. This makes a vector quite different from a scalar, which has only one characteristic, namely magnitude. Furthermore, there are two types of vectors: *polar vectors* and *axial vectors*.

Polar and axial vectors can be defined by the transformation rules they obey. Consider a coordinate transformation described by the 3×3 matrix a_{ij} ; we will restrict this matrix to represent a symmetry operator. A polar vector $\mathbf{E}$ is then defined as a mathematical quantity that satisfies the following transformation rule when its components are transformed from an old to a new reference frame (asterisks are used for components in the new reference frame):

$$E_i^* = a_{ij}E_j \quad (\text{polar}). \quad (8.19)$$

For an axial (or pseudo-) vector, $\mathbf{H}$, we must multiply this expression by the determinant of the transformation matrix:

$$H_i^* = \det |a| a_{ij} H_j \quad (\text{axial}). \quad (8.20)$$

The determinant of the matrix representing the symmetry operator is known as the *parity* of the operator, and is denoted by $p_a = \det |a|$.

Let us analyze the meaning of these transformation rules by means of a graphical example. Consider a single mirror plane, m , oriented normal to the $\mathbf{e}_z$ -axis through the origin. A polar vector $\mathbf{E}$ has components $[0, \cos \theta, \sin \theta]$ where $0^\circ \leq \theta \leq 90^\circ$, i.e., the vector is oriented parallel to $\mathbf{e}_y$ when $\theta = 0^\circ$ and is perpendicular to the mirror plane when $\theta = 90^\circ$.

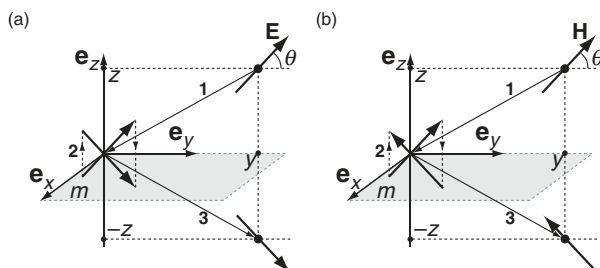


Figure 8.9 Illustration of the operation of a mirror plane m , parallel to the $x - y$ plane on a polar vector $\mathbf{E}$ (a), and an axial vector $\mathbf{H}$ (b). The operation can be broken down into three steps (denoted 1, 2, and 3).

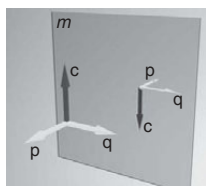


Figure 8.10 Mirror image of a right-handed triad $\mathbf{p}$, $\mathbf{q}$, $\mathbf{c} = \mathbf{p} \times \mathbf{q}$.

Let us also assume that this vector is located at the point with coordinates $\mathbf{r} = (0, y, z)$. The transformation of a vector under the symmetry operation m then consists of two steps: first, compute the image of $\mathbf{r}$ due to m , which is $(0, y, -z)$. Then move the vector $\mathbf{E}$ from $(0, y, z)$ to the origin, apply Eq. 8.19, and move the resulting vector to the point $(0, y, -z)$. Application of the mirror to the vector $\mathbf{E}$ is easily accomplished by matrix multiplication and results in the components $[0, \cos \theta, -\sin \theta]$. The steps are illustrated graphically in Fig. 8.9(a).

For an axial vector, $\mathbf{H}$, the same sequence of steps must be followed, with one exception: the transformation of the vector components must make use of Eq. 8.20 instead of 8.19. The parity, p_m , of the mirror operation equals -1 , so that the vector components of the mirror image of an axial vector must be reversed; in other words, the components of the mirror image of $\mathbf{H}$ are $[0, -\cos \theta, \sin \theta]$. The result, shown in Fig. 8.9(b), is somewhat surprising and counterintuitive, but it can be understood by considering Fig. 8.10. Two vectors $\mathbf{p}$ and $\mathbf{q}$ are shown, along with their cross product $\mathbf{c} = \mathbf{p} \times \mathbf{q}$. Recall that the direction of the cross product vector can be determined by application of the right-hand rule: make a triad with the first three fingers (thumb, index, and middle finger) of your right hand, orient your thumb along the first vector, $\mathbf{p}$, and your index finger along the second vector, $\mathbf{q}$; the cross product vector, $\mathbf{c}$, then points along your middle finger. This triad of vectors (or fingers) is hence known as a right-handed triad. When we look at the triad in a mirror, something interesting happens: the mirror images of $\mathbf{p}$ and $\mathbf{q}$ are determined easily, and when we apply the right-hand rule to obtain the cross product of these mirror

8.3 Magnetic symmetry operations

images, we find that the resulting vector points in the direction opposite to that of the vector $\mathbf{c}$. The only way in which the mirror image of $\mathbf{c}$ can be made to lie along the same direction as $\mathbf{c}$ is by applying an additional operation to flip the image vector around. This additional “flipping” operation is precisely the time-reversal operation. The same is true for Fig. 8.9(b): to make the mirror image of $\mathbf{H}$ point in the same direction as the mirror image of $\mathbf{E}$, we must apply an additional time-reversing operation. We leave it to the reader to determine the mirror image of the vector $\mathbf{H}$ in Fig. 8.9(b) for the special cases $\theta = 0^\circ$ and $\theta = 90^\circ$.

What we learn from this discussion is that, when applying symmetry operators, polar vectors behave as expected intuitively, but axial vectors will have the opposite direction if the symmetry operator has negative parity. As a consequence, polar vectors are invariant under the time-reversal operation; axial vectors, on the other hand, are reversed by the time reversal operator, so that the description of magnetic symmetry must include the time reversal operator as a regular symmetry operator. Since the combination of two symmetry operations is once again a symmetry operation, we will, in the next section, describe the time-reversal equivalents of the pure operations of the first and second kind.

8.3.2

Time-reversing symmetry operations

In most atoms, the orbital motions of electrons are such that they are paired into orbits with opposite moment vectors, so that the net moment vanishes; such atoms are non-magnetic, and they are typically represented by tiny spheres. In a small number of atoms (e.g., Fe, Ni, Co, ...) the orbital moments of the individual electrons do not fully cancel each other, so that these atoms have a net moment; they are magnetic atoms, and they must be represented as vectors that indicate the spatial orientation of the moment. When we operate on a non-magnetic atom at position $(x, y, 0)$ with a four-fold rotation around the z -axis, then we simply copy the tiny sphere representing the atom onto the equivalent positions generated by the rotation operator, i.e., the positions $(-y, x, 0)$, $(-x, -y, 0)$, and $(y, -x, 0)$. If the first atom is magnetic, with its moment vector parallel to the rotation axis, then we must allow for the possibility that the rotation may reverse the moment vector as well as rotate to the new position. While the regular four-fold rotation operation is represented by the symbol 4 (C_4), the time-reversing operation is represented by $4'$ (C'_4). In a single operation, the operator $4'$ (C'_4) rotates the atom by 90° around the axis *and* reverses the moment vector, as illustrated in 8.11(a). Note that the direction of the moment vector is arbitrary; in particular, it need not be parallel to the rotation axis at all. Two more examples of the application of $4'$ (C'_4) on two different moment orientations are shown in 8.11(b) and (c). This illustrates that a single 2-D drawing may not be sufficient to truly understand how a time-reversing symmetry operator works. Computer animation then becomes a useful tool to visualize the 3-D nature of the symmetry operator.⁹

In general, time-reversing rotations are indicated by the symbol n' (C'_n). It is important to note that not every time-reversing rotation produces a configuration of black and white

⁹ We refer the interested reader to the website <http://mpg.web.cmu.edu/>, which provides several 3-D animations and stills that illustrate how magnetic symmetry operators work.

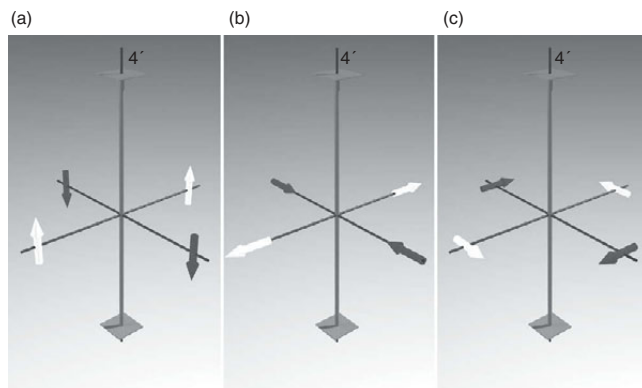


Figure 8.11 Illustration of the operation of $4'$ (C_4') on a magnetic moment vector, for three different orientations of the vector. The dark-gray vectors have been time-reversed with respect to the white vectors.

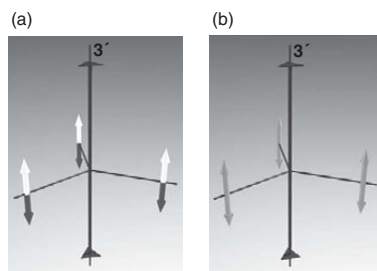


Figure 8.12 (a) Operation of the $3'$ (C_3') time-reversing rotation requires going around the axis twice. As a result, the operator produces a gray (non-magnetic) object at each equivalent location (b).

vectors, as in Fig. 8.11. In fact, any operator n' (C_n') of odd order produces a configuration in which a black and a white vector occupy the same locations, as shown in Fig. 8.12(a) for the operator $3'$ (C_3'). To return to the original location with the vector pointing in the same direction as the starting vector, we must perform two full rotations around the axis, which produces six vectors (three white, three black) at three equivalent positions. Since there are two opposite vectors at each location, there can be no net magnetic moment; we refer to the operator $3'$ (C_3') as a *gray* operator, as opposed to the two-color or black-and-white operator $4'$ (C_4').

The inversion operator $\bar{1}$ (i) also has a time-reversal equivalent denoted by $\bar{1}'$ (i'). Whereas a polar vector and its inverse point in opposite directions, an axial vector and its inverse point in the same direction, as shown in Fig. 8.13(a) and (b). It is easy to see that this is the case by considering a triad of vectors $\mathbf{c} = \mathbf{p} \times \mathbf{q}$, as in Fig. 8.10. In Fig. 8.13(c), the result

8.4 Combinations of symmetry operations

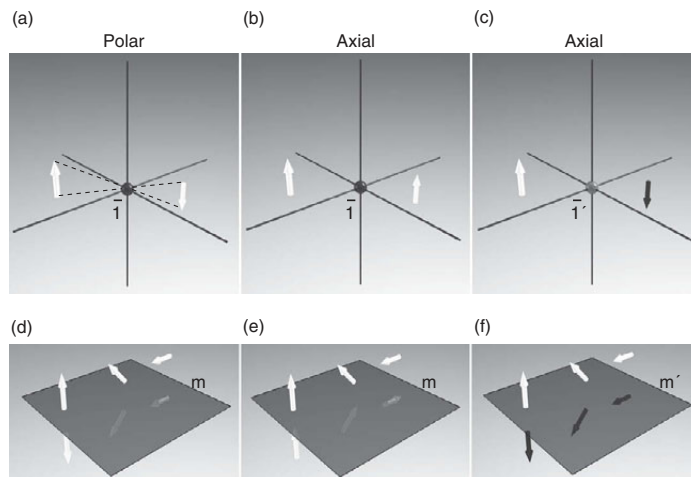


Figure 8.13 Illustration of the operation of $\bar{1}$ (i) (top row) and m (σ) (bottom row) on (a,d) a polar vector, and (b,e) an axial vector; (c,f) shows the operation of the time-reversed inversion $\bar{1}'$ (i') and m' (σ') on an axial vector.

of a time-reversed inversion $\bar{1}'$ (i') is shown; the time-reversed (black) vector now points in the same direction as the inverted polar vector of Fig. 8.13(a).

The last operator to be considered in this section is the time-reversing mirror operator m' (σ'). We have already seen that a regular mirror plane acts differently on polar and axial vectors; this difference is illustrated in Fig. 8.13(d) and (e) for different orientations of the vectors. When time-reversal is added to the mirror operation, the mirrored vectors have the same orientation as for the polar vector, as shown in Fig. 8.13(f).

8.4

Combinations of symmetry operations

8.4.1

Combination of rotations with the inversion center

The combination of a rotation axis with an inversion center located somewhere on that axis is called a *roto-inversion* operation. As an example, we combine a three-fold rotation 3 (C_3) with an inversion center $\bar{1}$ (i), as shown in the stereographic projections of Fig. 8.14. On the left-hand side, we show the projection of the standard three-fold rotation. On the right-hand side, we show the roto-inversion. The roto-inversion rotates a point over an angle $2\pi/3$ and inverts the resulting point through the inversion center, as shown in the middle projection.

The standard notation for a roto-inversion axis of order n is the symbol $\bar{n}$, e.g., $\bar{3}$ (the Schoenflies notation will be introduced in the following subsection). The roto-inversions

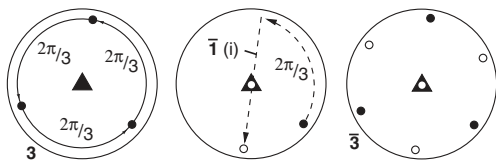


Figure 8.14 Regular three-fold rotation axis, operation of a three-fold roto-inversion and result of the repeated operation of the roto-inversion $\bar{3}$.

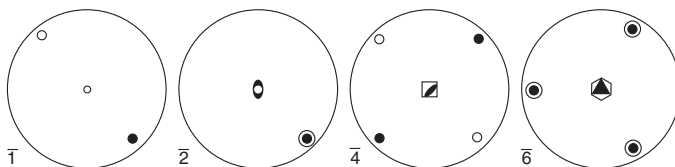


Figure 8.15 Stereographic projections for the roto-inversions $\bar{1}$, $\bar{2}$, $\bar{4}$, and $\bar{6}$.

relevant to crystallography are $\bar{1}$, $\bar{2}$, $\bar{3}$, $\bar{4}$, and $\bar{6}$. The corresponding stereographic projections are shown in Fig. 8.15. Note that for all values of n that can be written as $n = 4k + 2$, the inversion actually generates a mirror plane perpendicular to the rotation axis and the effective order of the rotation axis is only $n/2$. In other words, the roto-inversion $\bar{6}$ is equivalent to the rotation 3 (C_3) with a perpendicular mirror plane (σ_h). Note also that the two-fold roto-inversion $\bar{2}$ is equivalent to a simple mirror plane (σ). On drawings, the roto-inversions are represented by special symbols: $\circ$ for $\bar{1}$, $\dagger$ for $\bar{2}$, $\blacktriangle$ for $\bar{3}$, $\blacklozenge$ for $\bar{4}$, and $\blacklozenge$ for $\bar{6}$.

The transformation matrix associated with a roto-inversion is obtained by simply multiplying the rotation matrix with the matrix representing the inversion operator. As an example, consider the four-fold roto-inversion $\bar{4}$ oriented along the z -axis of a Cartesian reference frame:

$$\begin{aligned}
 \mathcal{W}_{\blacklozenge} &= \mathcal{W}_{\circ} \mathcal{W}_{\blacklozenge} \\
 &= \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{8.21}
 \end{aligned}$$

A point with coordinates (x, y, z) is hence moved to the position $(y, -x, -z)$.

8.4 Combinations of symmetry operations

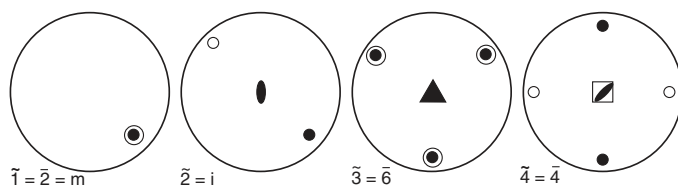


Figure 8.16 Stereographic projections for the mirror-rotations $\tilde{1}$, $\tilde{2}$, $\tilde{3}$, and $\tilde{4}$.

8.4.2

Combination of rotations and mirrors

A *mirror-rotation* is the combination of a rotation axis with a perpendicular mirror plane. Each point is first rotated over an angle $2\pi/n$, and the resulting point is then mirrored through the plane. We have already encountered an example in the previous section. Mirror-rotations of order n are usually indicated with the symbol $\tilde{n}$. Note that for odd values of n , the symmetry element $\tilde{n}$ is actually equivalent to $\overline{2n}$; for values of $n = 4N$, with N an integer, there is no difference between the symbol $\tilde{n}$ and $\bar{n}$, as is easily verified with a stereographic projection (reader exercise). Several mirror-rotations relevant to crystallography are shown in Fig. 8.16. For historical reasons, the Schoenflies notation makes use of mirror-rotations, while the Hermann–Mauguin notation uses the roto-inversions, so that the combined symbol is a little more difficult to remember. We introduce the following new Schoenflies symbol for the mirror-rotation:

$$S_n \equiv \sigma_h C_n,$$

where the h subscript on the mirror plane indicates that the plane is *horizontal*, i.e., perpendicular to the rotation axis. From these general rules and the illustrations in Figs. 8.15 and 8.16, it is not difficult to derive the list of relations shown in the Table 8.1. While the Schoenflies notation for these symmetry elements (third column) is easy enough to remember, it is, unfortunately, *not* the standard notation listed in the *International Tables for Crystallography, Volume A* (Hahn, 1996). The standard Schoenflies notation is listed in the last column of Table 8.1. The general rule used to construct this table takes the order of the roto-inversion axis, n , and considers three distinct cases:

- $n = 4N$, with N an integer: in these cases, there is no difference between the roto-inversion and the mirror-rotation. The official Schoenflies symbol is then S_n , and the complete symbol is denoted by $\bar{n} (S_n)$.
- $n = 2N + 1$ (i.e., n is odd): in all these cases, the equivalent mirror-rotation has *twice* the order of the roto-inversion, i.e., $\tilde{n} \equiv 2\bar{n}$. The official Schoenflies symbol is C_{ni} , so that the complete symbol becomes $\bar{n} (C_{ni})$ for n odd.
- $n = 4N + 2$ (i.e., even numbers that are not a multiple of four): in such cases, the equivalent mirror-rotation has *half* the order of the roto-inversion axis, i.e., $\tilde{n} \equiv \frac{1}{2}\bar{n}$. It is easy to show, using stereographic projections, that we have $\tilde{n} = \frac{1}{2}n/m$ (where the symbol n/m means that there is a mirror plane normal to the rotation axis). The official Schoenflies symbol is $C_{\frac{1}{2}nh}$, so that the complete symbol becomes $\frac{1}{2}n/m (C_{\frac{1}{2}nh})$.

Table 8.1 Hermann–Mauguin and Schoenflies notations for roto-inversions and mirror-notations

Hermann–Mauguin		S_n	Schoenflies
Roto-inversion	Mirror-rotation		
$\bar{1}$	$\bar{2}$	S_2	C_i
$\bar{2}(=m)$	$\bar{1}$	S_1	C_h
$\bar{3}$	$\bar{6}$	S_6	C_{3i}
$\bar{4}$	$\bar{4}$	S_4	S_4
$\bar{5}$	$\bar{10}$	S_{10}	C_{5i}
$\bar{6}(=3/m)$	$\bar{3}$	S_3	C_{3h}
$\bar{7}$	$\bar{14}$	S_{14}	C_{7i}
$\bar{8}$	$\bar{8}$	S_8	S_8
$\bar{9}$	$\bar{18}$	S_{18}	C_{9i}
$\bar{10}(=5/m)$	$\bar{5}$	S_5	C_{5h}
$\vdots$	$\vdots$	$\vdots$	$\vdots$

The 4-D transformation matrices associated with the mirror-rotations can be obtained easily by multiplying the matrix for a mirror plane by the rotation matrix (reader exercise). In the remainder of this book, we will always work with roto-inversions instead of mirror-rotations. However, the Schoenflies symbol for these operations will always be based on the mirror-rotation.

8.4.3

Combination of rotations and translations

It is a general property of the symmetry operations we have discussed so far that, with the exception of the *translation*, they all return to the same point when applied repeatedly. For instance, when we apply a six-fold rotation six consecutive times, we recover the initial point; when we apply a mirror operation twice, we recover the same point. This means that, for each symmetry element discussed so far, there exists an integer k , such that $\mathcal{O}^k = 1$ (E). If we include translations, then this statement is no longer true, since repeated operation of the translation operator takes us ever further away from the initial point.

Combining translations with rotations yields a new type of symmetry element: the *screw axis*. Consider a rotation axis of order n , and apply it several times in a row. After every rotation step, we also *translate* the resulting point by a certain vector τ , parallel to the rotation axis. If we apply this operation n times, then the resulting point will be translated with respect to the original point by a vector $n\tau$. Since we are working in a 3-D lattice, this new point must again be a lattice point, which means that $n\tau = m\mathbf{t}$, where $\mathbf{t}$ is the shortest lattice vector parallel to the rotation axis. We have, hence, created a new symmetry operation, the *screw axis*, which consists of an n -fold rotation combined with a translation parallel to the rotation axis $\mathbf{t}$ by a vector $\tau = m\mathbf{t}/n$. The distance $\tau = |\tau|$ is called the *pitch* of the screw axis. The resulting symmetry element is generally denoted by the symbol $\mathbf{n}_m$. If $m = 0$, then there is no translation associated with the rotation and

8.4 Combinations of symmetry operations

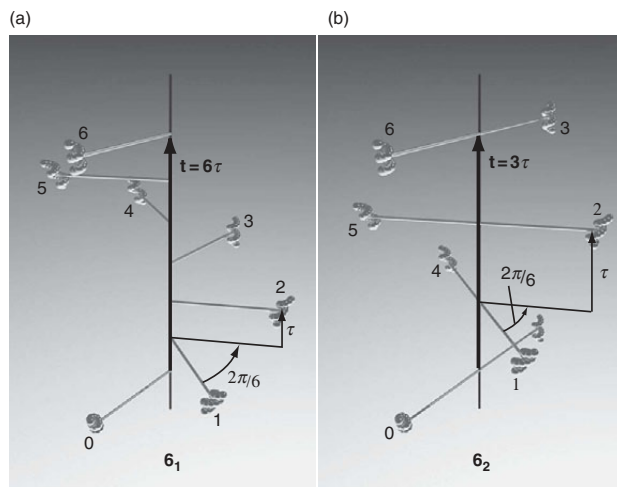


Figure 8.17 Rendered perspective drawings of the operations involved in the construction of the screw axes 6_1 (a) and 6_2 (b).

we recover the proper rotations as a subset of the screw rotations. The value of m must lie between 0 and n .

An example of the screw axis 6_1 is shown in Fig. 8.17(a): the vertical line indicates the screw axis. The helix labeled 0 is the starting point. After a rotation of $2\pi/6$, the resulting helix is translated by a vector $\tau = \mathbf{t}/6$ to the point 1. Another rotation followed by a translation leads to point 2 and so on, until point 6, which is located at a lattice vector $\mathbf{t} = 6\tau$ from point 0. The 6_2 operation, shown in Fig. 8.17(b), consists of a 60° rotation and a translation by $\tau = 2\mathbf{t}/6 = \mathbf{t}/3$. Both vectors $\mathbf{t}$ and τ are indicated on the drawing. The helix at point 0 is first rotated over 60° counterclockwise (indicated by the curved arrow), then translated over τ , rotated again, translated, etc. After three rotations, the resulting point has been translated by a distance $3\tau = \mathbf{t}$. The following three points are located in the next unit cell along the rotation axis. Since $\mathbf{t}$ is a lattice vector, there must also be corresponding points in the lower unit cell (points 4, 5, and 6). We can construct the same drawing for the 6_4 -axis (reader exercise); the resulting screw is identical to 6_2 , but has opposite rotation direction. We thus find that 6_4 produces the mirror image of 6_2 . This is true for all screw axes: the screw axes of the type n_n and n_{n-m} are each other's mirror image. A screw axis is called *right-handed* if $m < n/2$, *left-handed* if $m > n/2$ and *without hand* if $m = 0$ or $m = n/2$. Screw axes related to each other by a mirror operation are called *enantiomorphous*.

We have seen in Section 8.2.2 that the only rotations of importance to classical crystallography are two-, three-, four-, and six-fold rotations; hence, the only screw axes that we will consider in this text are (with their official graphical symbols): 2_1 †, 3_1 ▲, 3_2 ▲, 4_1 ◆, 4_2 ◆, 4_3 ◆, 6_1 ★, 6_2 ★, 6_3 ★, 6_4 ★, and 6_5 ★. All screw axes are shown in Fig. 8.18 as both rendered drawings and standard graphical representations: the number

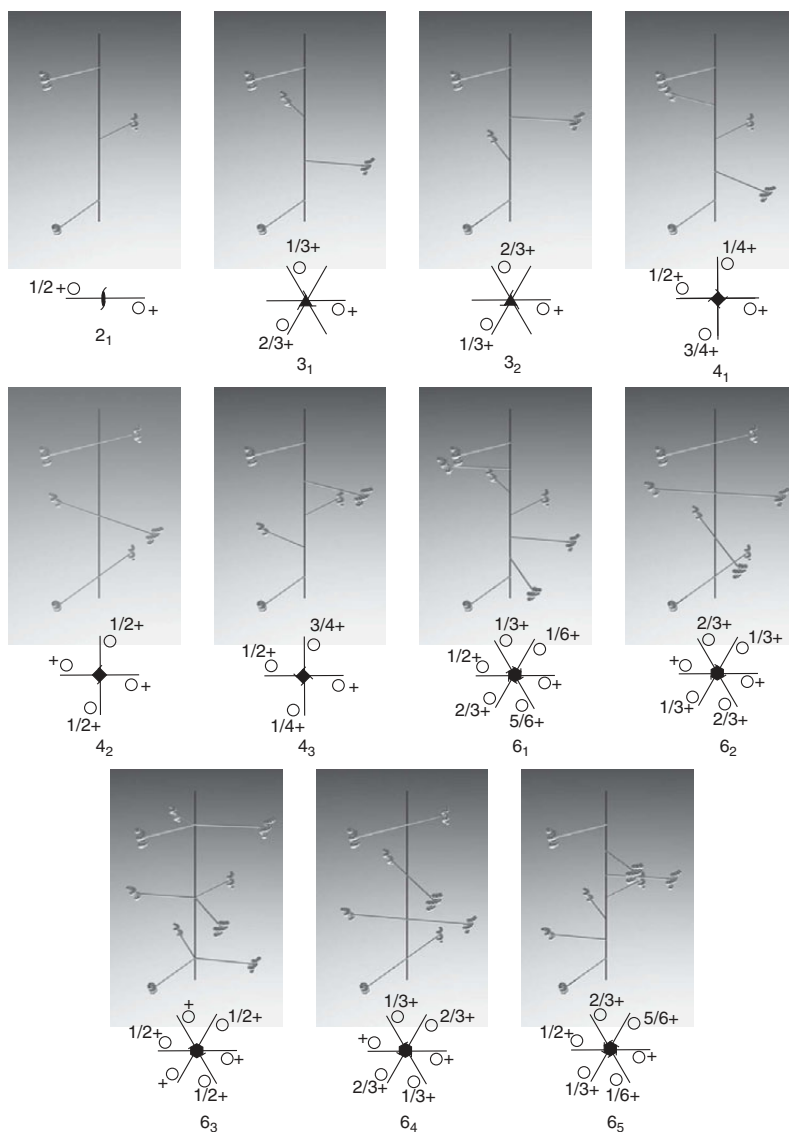


Figure 8.18 Rendered perspective drawings of the crystallographic screw axes.

8.4 Combinations of symmetry operations

next to each circle refers to the height of the circle above the plane of the drawing. The axes are all perpendicular to the drawing.

To conclude this section on screw axes, we determine how we can represent this symmetry operator by means of a matrix. We have already seen all the ingredients that we need to come up with a compact notation. The screw axis has both a rotation and a translation, so that the Seitz symbol is given by $(D(\theta)|\tau)$. The 4×4 matrix $\mathcal{W}$ is then easily constructed by taking the 3×3 rotation matrix and combining it with the column vector of the translation components. As an example, a screw axis of the type $\bullet$ along the c -axis of a hexagonal reference frame is described by the matrix:

$$\mathcal{W} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{6} \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (8.22)$$

We leave it as an exercise for the reader to determine the matrix representation of a 4_2 operator parallel to the c -axis of a tetragonal reference frame, when the screw axis goes through the point with coordinates $(1/4, 3/4, 0)$.

8.4.4

Combination of mirrors and translations

The last class of symmetry operators involves combinations of mirror planes and translations. As indicated in Fig. 8.19(a), one could combine a mirror plane with a translation over a *lattice vector* parallel to the plane. This does not give rise to a new symmetry operation. One then defines the *glide plane* as the combined operation of a mirror with a translation over *half* a lattice vector parallel to the mirror plane. An example is shown in Fig. 8.19(b).

The allowed glide vectors must be equal to one half of the lattice vectors. In the case of centered Bravais lattices, there may be additional glide vectors, equal to one half of the

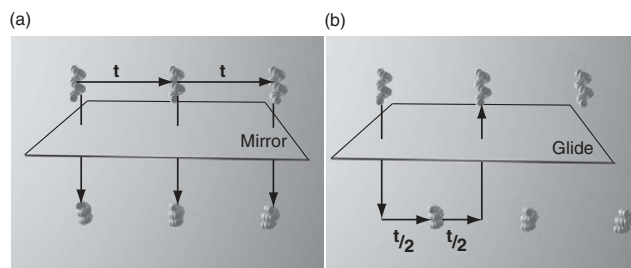


Figure 8.19 (a) The combination of a mirror plane with an integer-valued translation does not give rise to a new symmetry operation; (b) a glide plane, consisting of a mirror plane and a half-integer translation.

Table 8.2 Different types of mirror and glide planes, their symbols and glide vectors. Glide vectors with a $(\dagger)$ symbol are only possible in cubic and tetragonal systems.

Name	Symbol	Glide vector(s)
Mirror	m	None
Axial glide	a	$\mathbf{a}/2$
	b	$\mathbf{b}/2$
	c	$\mathbf{c}/2$
Diagonal glide	n	$\mathbf{A}, \mathbf{B}, \mathbf{C}$ $\mathbf{I} \quad (\dagger)$
Diamond glide	d	$(\mathbf{a} \pm \mathbf{b})/4, (\mathbf{b} \pm \mathbf{c})/4, (\mathbf{c} \pm \mathbf{a})/4$ $(\mathbf{a} \pm \mathbf{b} \pm \mathbf{c})/4 \quad (\dagger)$

centering vectors $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{I}$. Table 8.2 lists the various possibilities and names for all types of glide planes.¹⁰

The official drawing symbols for glide planes depend on the orientation of the plane with respect to the drawing. If the plane is perpendicular or at an angle to the drawing, then the plane is indicated by a *solid bold* line for m (σ), a *dashed bold* line for a glide with translation *in* the plane of the drawing, a *dotted bold* line for a glide with translation *perpendicular* to the plane of the drawing, and a *dash-dotted bold* line for a *diagonal* glide (see Fig. 8.20). Note that, once again, a circle reflected in a mirror plane is represented by a circle with a comma in the center, to indicate that an odd number of reflections relate that point to the original point. For glide planes parallel to the plane of the drawing, one uses a symbol based on $\neg$, which represents a pure mirror. For an *axial glide plane*, one adds an arrow to the symbol in the direction of the glide vector, i.e., $\neg\rightarrow$. For a *diagonal glide plane*, the arrow points at an angle away from the corner, as in $\neg\swarrow$. Examples are shown in the right side column of Fig. 8.20.

The matrix representation for a glide plane is readily obtained by combining the 3×3 mirror operation with the appropriate translation components. For instance, an a -glide reflection in the plane $z = 0$ converts the z -coordinate of any point into $-z$ and translates that point by $(1/2, 0, 0)$. The resulting matrix is hence given by:

$$\mathcal{W} = \begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (8.23)$$

¹⁰ In the *International Tables for Crystallography, Volume A* (Hahn, 1996), the concept of *double glides* is used. This refers to the fact that sometimes both a and b glides are simultaneously present. A special graphical symbol and notation are used to describe these cases (see page 6 of the *Tables*), but, in this book, we will not make any use of double glide planes.

8.4 Combinations of symmetry operations

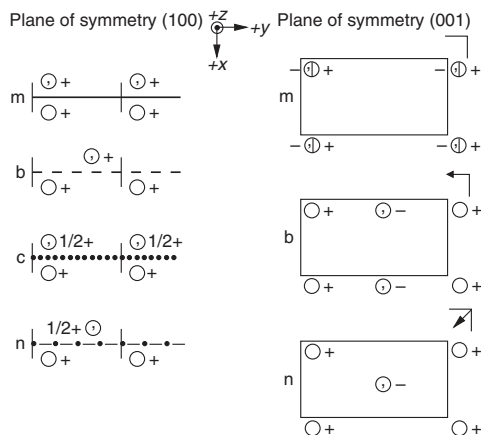


Figure 8.20 Graphical symbols for glide planes perpendicular (left) and parallel (right) to the plane of the drawing (Figure reproduced from Fig. 1.17 in *Introduction to Conventional Transmission Electron Microscopy*, M. De Graef, 2003, Cambridge University Press).

The Seitz symbol is also shown readily to be $(D(m)|\tau)$, where τ is the glide vector. We leave it to the reader to determine the transformation matrix $\mathcal{W}$ for a diagonal $\mathbf{B}$ glide reflection parallel to a (010) mirror plane going through the point $(1/2, 1/2, 1/2)$ (cubic reference frame).

8.4.5

Relationships and differences between operations of the first and second kind

In this section, we will take a closer look at the differences between operators of the first and second kind. From the examples in the previous sections, we can infer that the proper motions (operations of the first kind) are described by orthogonal matrices with positive unit determinant, i.e.,

$$\det |{}^I D| = +1, \quad (8.24)$$

whereas all transformations which change the handedness of an object (operations of the second kind) have

$$\det |{}^II D| = -1. \quad (8.25)$$

Successive products of any number q of operators of the first kind leave the determinant unchanged:

$$\det |{}^I D_q {}^I D_{q-1} \dots {}^I D_1| = +1. \quad (8.26)$$

However, for operators of the second kind, the handedness of the product is determined by the number of operations: if q is even, then the resulting operation is of the first kind,

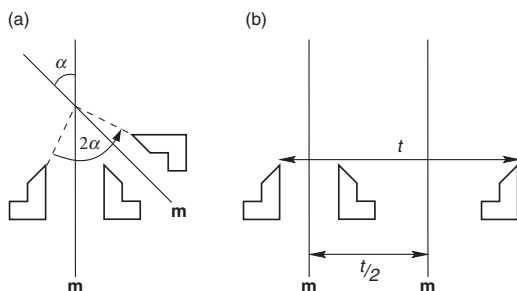


Figure 8.21 Combination of intersecting symmetry elements creates new elements, according to Euler's theorem. Some limiting cases for intersecting and parallel mirror planes are shown in (a) and (b), respectively.

otherwise it remains of the second kind. The determinant of the composite operation $C = {}^H D_q {}^H D_{q-1} \dots {}^H D_1$ equals

$$\det |C| = (-1)^q = \begin{cases} +1 & q = 2n \quad \rightarrow C \text{ is first kind operator,} \\ -1 & q = 2n + 1 \quad \rightarrow C \text{ is second kind operator.} \end{cases}$$

Since the product of two orthogonal matrices is again an orthogonal matrix, it follows that the combination of two symmetry operations must again be a symmetry operation. Figure 8.21 shows two special cases: on the left, two intersecting mirror planes are shown to be equivalent to a rotation over twice the angle between the mirror planes around their intersection line. On the right, it is shown that the combination of two parallel mirror planes with a spacing $t/2$ is equivalent to a translation over a distance t .

We thus find that intersecting mirror planes always imply the existence of rotational symmetry. Parallel mirror planes always imply the existence of translational symmetry. In general, it can be shown that the combination of two rotations about intersecting axes can always be replaced by a single rotation; this is known as *Euler's theorem* and the complete proof of this theorem can be found in McKie and McKie (1986), pages 43–47.¹¹ This theorem is important for the derivation of all the point groups in Chapter 9.

8.4.6

*Combinations of magnetic and regular symmetry operators

When the time-reversal operator $1'$ (E') is combined with the symmetry operators introduced in the previous section, a whole series of new, time-reversing operators is obtained. For each screw axis and glide plane, we must consider what happens when the operator also reverses time, in addition to rotating and translating, or mirroring and translating, respectively. In this section, we will only consider a few examples; for a complete listing

¹¹ Euler has many theorems that carry his name, and this is just one of them.

8.4 Combinations of symmetry operations

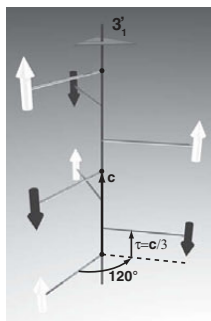


Figure 8.22 Illustration of a $3_1'$ time-reversing screw axis.

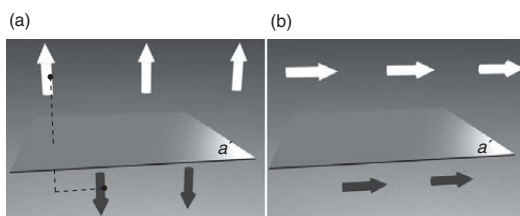


Figure 8.23 Illustration of the operation of a time-reversing glide a' on an axial vector that is (a) normal and (b) parallel to the glide plane.

of time-reversing operators and their graphical symbols we refer to pages 326–345 in Litvin (2010).¹²

The first example is the screw axis 3_1 , introduced on page 179. The time reversing equivalent of this operator is written as $3_1'$, and is illustrated in Fig. 8.22. The black and white arrows alternate along the axis, and the repeat distance is twice as long as for the 3_1 -axis. Other time-reversing screw axes, such as $4_3'$ and $6_2'$, can be visualized in a similar way. Note that the arrows in Fig. 8.22 represent axial vectors. We leave it to the reader to determine the configuration of white and black vectors for the case where the bottom white vector in Fig. 8.22 is normal to the screw axis instead of parallel.

Time-reversing glide planes can be defined in a similar way, by combining the time-reversing mirror, m' (σ'), with a glide vector. Once again we must take into account the orientation of the axial vector with respect to the glide plane. Two special cases are shown in Fig. 8.23 for the time-reversing a -glide, a' : in (a), the left-most axial vector (white) is normal to the mirror plane, and its regular mirror image, according to Fig. 8.13(e), points

¹² This reference is available as an electronic book via the following link: www.bk.psu.edu/faculty/Litvin/Download/V3.02_Magnetic_Space_Groups_Book.pdf; this PDF file contains an astounding 4472 pages with complete information on all 1651 magnetic space groups (see Chapter 10 for more information).

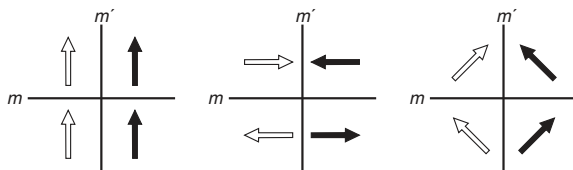


Figure 8.24 Intersection of a regular and a time-reversing mirror plane, along with several possible orientations of the mirror images of axial vectors.

in the same direction; hence, the time-reversed vector (black) points downwards. This pattern repeats itself along the **a** translation direction. When the first vector is parallel to the time-reversing glide plane, the operation of the glide a' results in a series of vectors, all pointing in the same direction, as shown in Fig. 8.23(b). We leave it to the reader to generate drawings of the other time-reversing glide planes based on Table 8.2.

We conclude this section by considering what happens when a regular and a time-reversing symmetry operator intersect each other. For instance, what happens when a regular mirror plane, $m(\sigma)$, is perpendicular to a time-reversing mirror, $m'(\sigma')$? Figure 8.24 illustrates this configuration for three different orientations of the initial axial vector. It is of interest to note that the total moment (sum of the moment vectors) is different in each of the three cases, varying from zero for the center configuration to four times the single moment for the one on the left. We leave it to the reader to determine what happens when one of the parallel mirror planes in Fig. 8.21(b) becomes a time-reversing mirror, and the object is an axial vector.

8.5

Point symmetry

It is instructive to take a closer look at the 3×3 sub-matrix of the 4×4 symmetry matrices. If we ignore all translations, then all symmetry matrices of the type:

$$\mathcal{M} = \begin{pmatrix} D_{11} & D_{12} & D_{13} & 0 \\ D_{21} & D_{22} & D_{23} & 0 \\ D_{31} & D_{32} & D_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

represent symmetry elements, which *all intersect each other in one common point, namely the origin*. This is easy to see: when **D** is multiplied by the position vector (0,0,0), the result is always (0,0,0), regardless of the particular details of **D**. Hence, the origin is the only point that is *invariant* under all symmetry operations **D**. For this reason, these operations are called *point symmetries*.

The question we must answer next is the following: is it possible for an object to have multiple rotational symmetries, and, if so, which combinations of rotation axes are allowed? The first part of this question is easy to answer. Take a cylinder, for example. The

8.5 Point symmetry

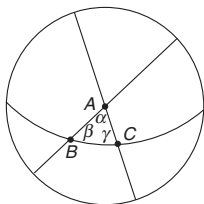


Figure 8.25 Three rotation axes A , B , and C on a stereographic projection, used to illustrate Euler's theorem.

cylinder has a rotation axis of infinite order parallel to its long axis, and an infinite number of two-fold rotation axes perpendicular to the long axis, going through the center plane. Thus, it is possible for an object to have multiple rotational symmetries.

In the case of crystals, we must refine the second part of the question to: which combinations of rotation axes are compatible with translational periodicity? In other words, under which conditions can two (or more) rotation axes with different orientations simultaneously be compatible with the Bravais lattice? Consider the stereographic projection in Fig. 8.25: there are three rotation axes, represented by the poles A , B , and C . Then Euler's theorem states (without proof): *If the angle between the great circles AB and AC is α , the angle between the great circles BA and BC is β , and the angle between the great circles CA and CB is γ , then a clockwise rotation about A through the angle 2α , followed by a clockwise rotation about B through the angle 2β is equivalent to a counterclockwise rotation about C through the angle 2γ .*

If the rotation axes at A , B , and C are n -fold rotation axes, then the angles 2α , 2β , and 2γ must be of the type $2\pi/n$ with, in the case of crystals, $n = 1, 2, 3, 4, 6$. We can turn this statement around so that the angles between the great circles connecting different rotation axes must be of the type $2\pi/2n$, or π/n . Note that these are angles between great circles, i.e., they are measured on the surface of the projection sphere. The angles between the rotation axes themselves can then be derived by means of spherical trigonometry. It can be shown that the angle between the rotation axes B and C is given by

$$\widehat{BC} = \arccos\left(\frac{\cos \alpha + \cos \beta \cos \gamma}{\sin \beta \sin \gamma}\right), \quad (8.27)$$

with similar relations for the other pairs of axes. If we substitute all possible combinations for α , β , and γ , and compute the angles $\widehat{AB}$, $\widehat{BC}$, and $\widehat{CA}$, then we arrive at the values shown in Table 8.3.

We find that only six different combinations of multiple rotation axes are compatible with the Bravais lattices. Those combinations are: 222, 223, 224, 226, 233, and 234. They are represented as rendered perspective drawings and stereographic projections in Fig. 8.26. Other combinations, such as $22n$ with $n = 5$ (also shown in the table) or $n > 6$, are possible for isolated objects, but are not compatible with the Bravais lattice translational symmetry. The combination 235 (last line in Table 8.3), known as the *icosahedral symmetry*, is shown as a rendered drawing in Fig. 8.27. This and other examples of non-crystallographic symmetries will be discussed in Chapter 15.

Table 8.3 Combinations of three rotation axes for crystallographic symmetries. A * indicates that the cosine of one of the angles falls outside the range $[-1, 1]$. The bottom two lines show combinations involving the non-crystallographic five-fold rotation axis; these will become important in Chapter 15.

n_A	n_B	n_C	$\widehat{BC}$	$\widehat{CA}$	$\widehat{AB}$	Possible?
2	2	2	90°	90°	90°	Yes
2	2	3	90°	90°	60°	Yes
2	2	4	90°	90°	45°	Yes
2	2	6	90°	90°	30°	Yes
2	3	3	$70^\circ 32'$	$54^\circ 44'$	$54^\circ 44'$	Yes
2	3	4	$54^\circ 44'$	45°	$35^\circ 16'$	Yes
2	3	6	0	0	0	Trivial
2	4	4	0	0	0	Trivial
2	4	6	*	*	*	No
2	6	6	*	*	*	No
3	3	3	0	0	0	Trivial
3	3	4	*	*	*	No
3	3	6	*	*	*	No
3	4	4	*	*	*	No
3	4	6	*	*	*	No
3	6	6	*	*	*	No
4	4	4	*	*	*	No
4	4	6	*	*	*	No
4	6	6	*	*	*	No
6	6	6	*	*	*	No
2	2	5	90°	90°	36°	Not in crystals
2	3	5	$37^\circ 23'$	$31^\circ 43'$	$20^\circ 54'$	Not in crystals

8.6

Chapter summary

In this rather long chapter, we have defined the concept of symmetry, as it applies to space lattices. We started with the four basic symmetry operators, translations, rotations, mirrors, and the inversion, and defined both graphically and mathematically how they work. We introduced two notational conventions: the international Hermann–Mauguin notation, and the Schoenflies notation. Then we looked at combinations of these operators, to form

8.6 Chapter summary

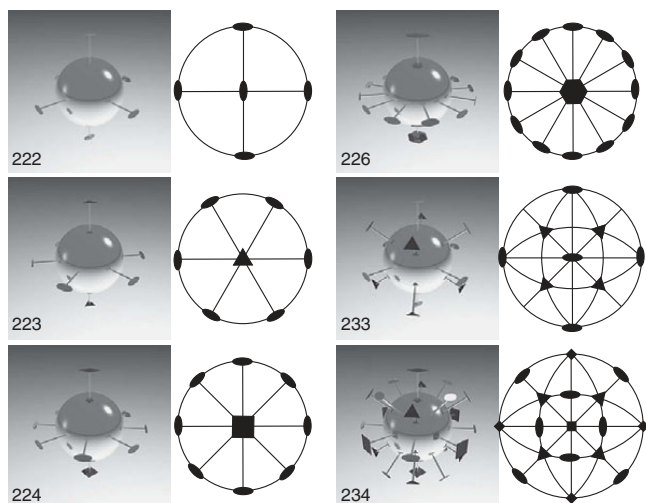


Figure 8.26 Rendered perspective drawings and stereographic projections showing all six crystallographic combinations of multiple rotation axes.

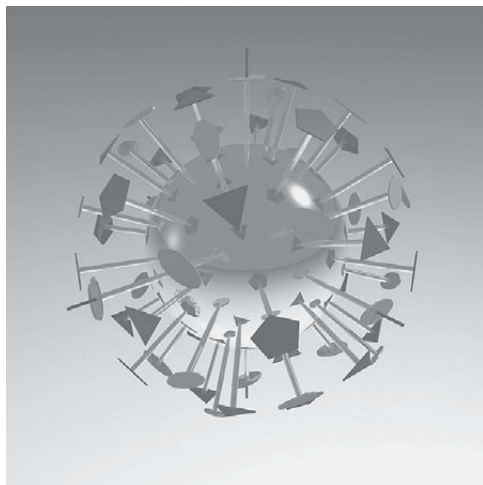


Figure 8.27 Rendered drawing of the icosahedral combination of rotation axes 235.

screw axes and glide planes. We also considered what happens when the crystal lattice is decorated with atoms that carry a magnetic moment. We defined axial and polar vectors, as well as the time-reversal operator, which leads to a new class of time-reversing symmetry operators. We concluded the chapter with the definition of point symmetry as well as the enumeration of the combinations of rotation axes that are compatible with the Bravais translational symmetries. In the next chapter, we will systematically work through the allowed combinations of point symmetry operators and derive the 32 crystallographic point groups and the 122 magnetic point groups. Adding translations to the point symmetry operators will then lead to the enumeration of the 230 3-D space groups in Chapter 10.

8.7

Historical notes

Arthur Moritz Schoenflies (1853–1928) was a German mathematician. He studied at the University of Berlin from 1870 until 1875 and subsequently received a doctorate from Berlin in 1877. Schoenflies (Fig. 8.28(a)) was appointed as chair of applied mathematics at Göttingen, set up by Felix Klein in 1892. In 1899, he was named chair at Königsberg. In 1911, he became Professor at the Academy for Social and Commercial Sciences in Frankfurt. Schoenflies served as professor from 1914 until 1922 at the University of Frankfurt and was rector during the period 1920–21.

Schoenflies was one of three men who, in the 1890s, almost simultaneously derived the existence of the 230 space groups (which will be discussed in Chapter 10). Schoenflies also developed the famous Schoenflies notation for the crystal forms index. He applied his space group theory to crystal structures and to the problem of the division of space into

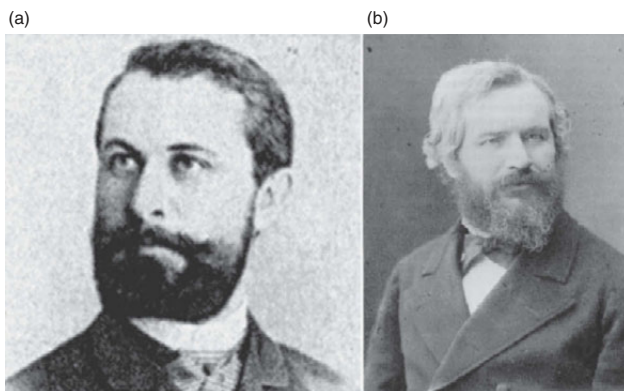


Figure 8.28 (a) Arthur Moritz Schoenflies (1853–1928) (picture courtesy of J. Lima-de-Faria), and (b) Camille Jordan (1838–1922) (picture courtesy of J. Lima-de-Faria).

8.8 Selected problems

congruent regions related by the symmetry of the group. His interest in symmetry guided many of his scientific endeavors.

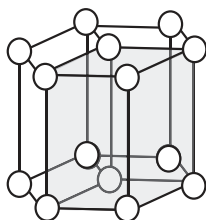
Many of the initial efforts in describing the way that atoms and molecules are arranged in crystals were developed in the eighteenth, nineteenth, and twentieth centuries prior to the development of techniques that actually probe the atomic structure. The simple packing models were precursors of the monumental developments that occurred as a result of considerations of symmetry and the mathematical properties of groups (in this case groups of symmetry operations). In 1844, Cauchy studied the group properties of permutations, a mathematical construct that was a precursor to symmetry group operations that will be discussed in subsequent chapters.

Mathematicians such as **Christian Wiener** (1826–96) and **Camille Jordan** (1838–1922) contributed to the development of the theory of space groups. Wiener developed a theory of *symmetric repetitions*, and Jordan (Fig. 8.28(b)) studied *groups of motions*. Jordan's work on group theory was published in the first ever group theory text book: *Traité des substitutions and des équations algébrique* (1870). **Christian Felix Klein** (1849–1925), an influential German mathematician, studied the problem of space groups from the point of view of transformation groups, extending the work of Jordan to consider adding improper rotations to the discrete group of proper rotations (1892). He also published the important work, "Lectures on the icosahedron" in 1876, in which he laid out the group theory for icosahedral groups. The German mathematician **Georg Frobenius** (1849–1917) was also known for his work in group theory. His famous congruence is the basis for an algorithm for the construction of the space groups.

8.8

Selected problems

- (i) *Hexagonal crystal system*: Consider the hexagonal prismatic representation of three unit cells illustrated in Fig. 8.29. Give examples of six-fold, three-fold, and two-fold rotational symmetries and a mirror plane that leaves the infinite lattice invariant (i.e., show axes about which you can rotate the infinite lattice and not tell it apart). Comment on the usefulness of the hexagonal prismatic representation in illustrating these symmetries as opposed to a single unit cell.



Hexagonal (hP)

Figure 8.29 Hexagonal unit cells.

- (ii) *Permutations*: Proceeding counterclockwise, label the vertices of a square with the labels 1, 2, 3 and 4, respectively.
 - (a) Determine all of the permutations of the numbers, 1, 2, 3 and 4.
 - (b) With respect to the square, determine the symmetry operation that is associated with each permutation.
 - (c) Write down a 2-D matrix representing each of these symmetry operations.
- (iii) *4-D transformation matrices*:
 - (a) Determine the Seitz matrix representation $\mathcal{W}$ of a 3_2 operator parallel to the **c**-axis of a rhombohedral reference frame.
 - (b) Determine the Seitz matrix, $\mathcal{W}$, for a mirror operation through the (11.0) plane going through the origin.
 - (c) Determine the coordinates of a point (x, y, z) , after it has undergone transformation by a screw axis 6_3 along the hexagonal **c**-axis, followed by a mirror operation through the (11.0) plane going through the origin.
- (iv) *Magnetic symmetry operators*: Make a sketch of each of the magnetic screw axes $4'_3$ and $6'_2$; write down the transformation matrix for each operator and indicate what the repeat distance is along the screw axis (with respect to the non-time-reversing version of each operator).